Lecture 1

SWE 4101

Introduction to Software Engineering

Tanjila Alam Sathi
Lecturer,
Department of CSE,
Islamic University of Technology

Outline

- Computer
- History of Computation
- Data Processing
- Components of a Computer System

Computer

- Early days the computer was not a machine !!!

- The word Computer comes from the word “Compute”.
- Thereby, a computer is a device that can perform arithmetic and logical operation at high speeds.
- A computer is also known as data processor.

History of Computation



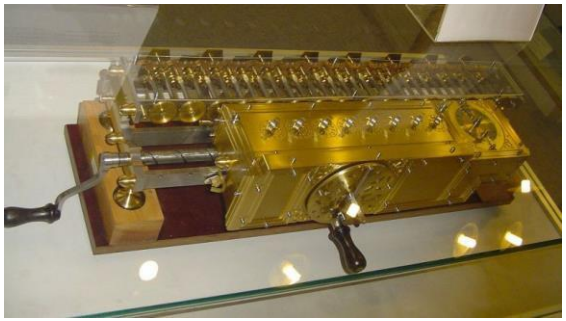
The Chinese Abacus 'Suan Pan'



Pascal's Calculator



The Roman Abacus



Step Reckoner

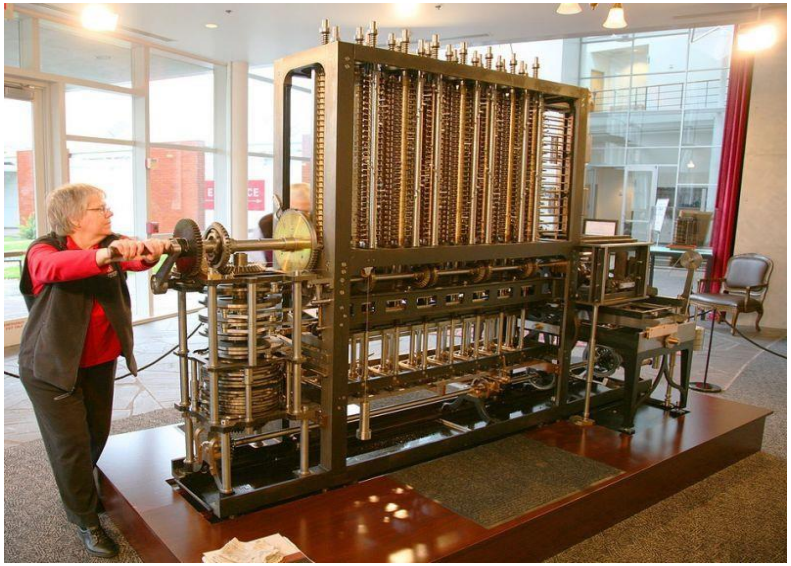
History of Computer

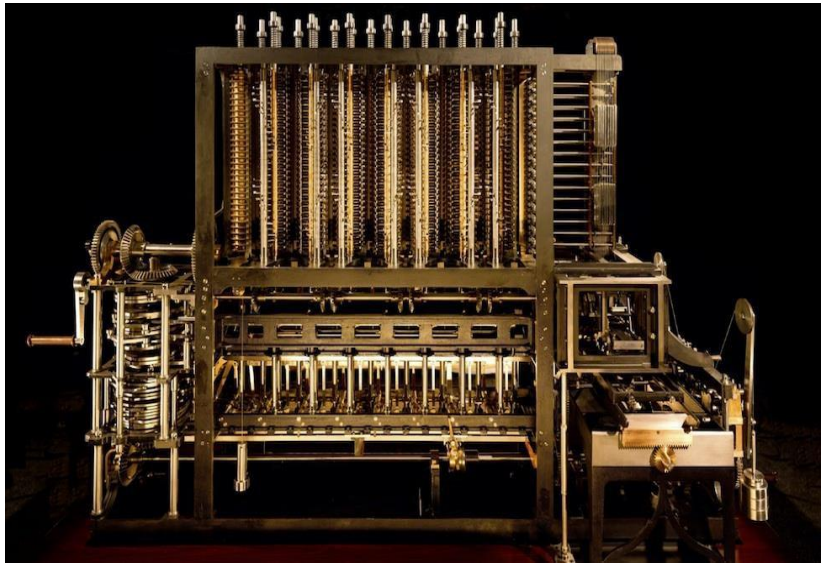




Punched Card System

History of Computer



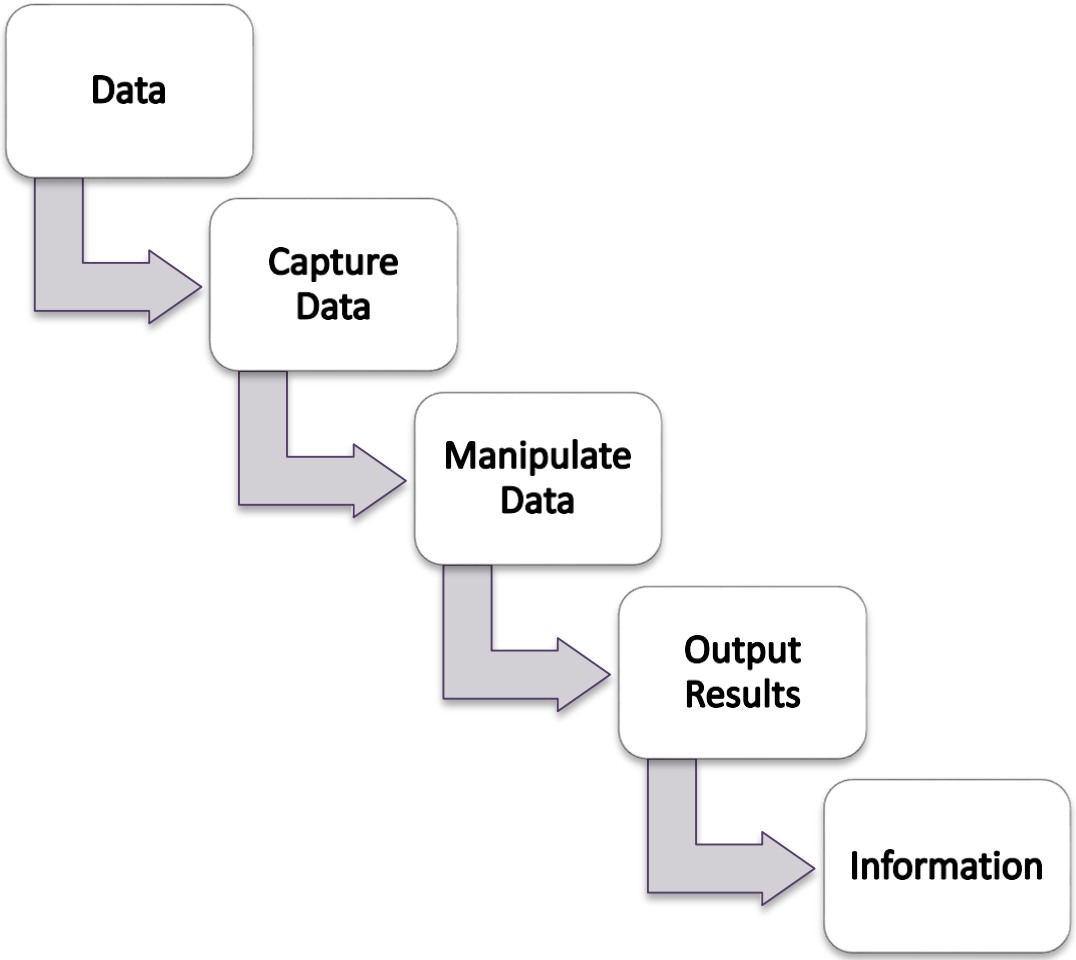


The Difference engine

History of Computer

- Modern Computer
  - Turing Machine
  - Universal Turing Machine
  - Electromechanical Computer
  - Stored-program Computers.

Data Processing



Components of a Computer System

- A computer system consists of –
  - Hardware - The **hardware** components of a computer system are the electronic and mechanical parts.
  - Software - The **software** components of a computer system are the data and the computer programs

Hardware Components

- Processor
- Main memory
- Secondary memory
- Input devices
- Output devices

Reading Materials

- Computer Fundamentals (by Anita Goel)
  - Chapter 1